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INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Project No. 2 - New Car Market Analysis

About me

- **Background? (B-tech or M-tech)**
- **Name:** Shriya Mechineni
  Education: B.E./B.Tech in Computer Science Engineering
  College: Prof. Ram Meghe College of Engineering & Management (Passout)

- **Why you want to learn Data Science**
- I want to learn Data Science because I enjoy working with data and finding useful patterns from it. I want to develop practical skills in Python, SQL, data analysis, visualization, and machine learning and use them to solve real-world problems.

- **Any work experience**
- No professional work experience

- **Share your LinkedIn and GitHub profile URLs**
- https://www.linkedin.com/in/shriya-mechineni-74851331a?utm_source=share_via&utm_content=profile&utm_medium=member_android

Introduction

- **Project Title: New Car Market Analysis**
- This project focuses on collecting and analyzing Hyundai car information from the CarWale website.
- The project demonstrates the complete data analytics workflow:
- **Web Scraping → Data Collection → Data Cleaning → EDA → Analysis → Visualization**
- The dataset contains information such as price, fuel type, transmission, engine capacity, mileage, seating capacity, and user rating.

Business Problem Statement

- Car buyers need to compare different cars based on factors such as:

1. Price
2. Fuel type
3. Transmission
4. Engine capacity

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5. Mileage

6. Seating capacity

7. User rating

- Manually collecting and comparing this information from a website can be time-consuming.
- Therefore, this project aims to automate data collection and analyze the available Hyundai car information using Python.

Project Objective

- The main objectives of this project are:
- Extract Hyundai car information from CarWale.
- Store the collected information in a structured dataset.
- Clean and preprocess the data.
- Analyze price, fuel type, transmission, mileage, engine capacity, and ratings.
- Identify budget, mid-range, and premium cars based on price.
- Create visualizations to understand market patterns.
- Demonstrate an end-to-end data analytics workflow

Website Selection & Data Source

Website: CarWale

Website URL:
https://www.carwale.com/

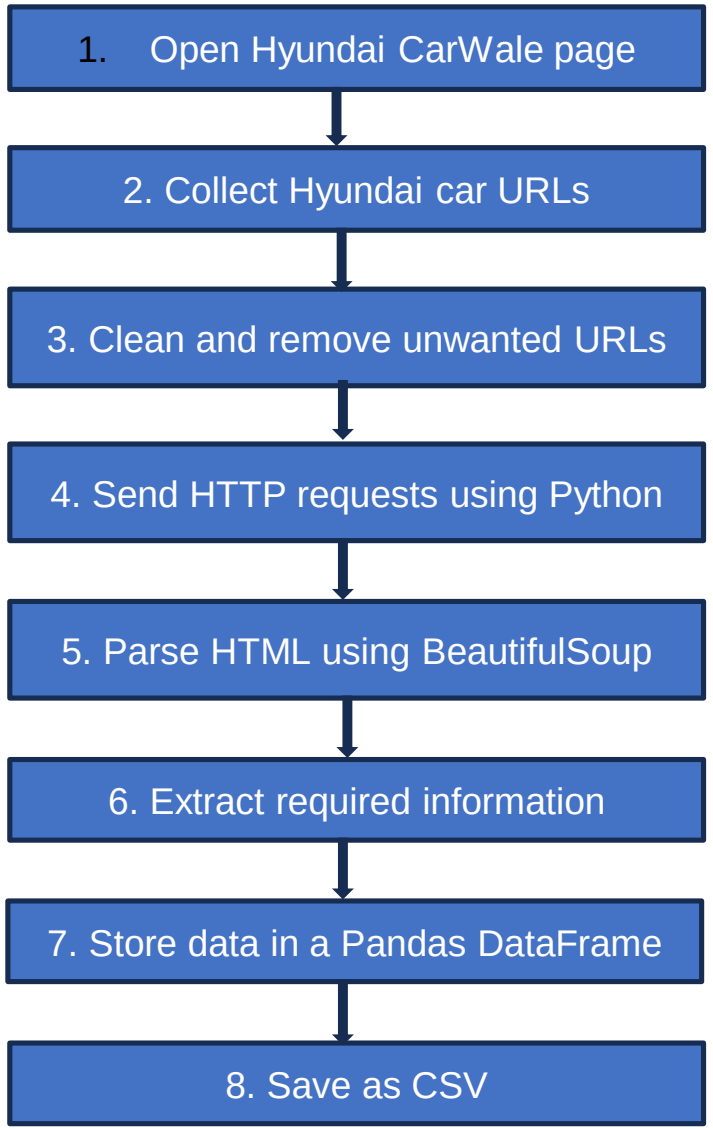
Data Source: Hyundai car model pages available on CarWale.

Reason for selection:
Contains car-specific information.
Provides details such as price and specifications.
Multiple Hyundai model pages are available.
Suitable for practicing web scraping and data analysis.

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* Web Scraping Methodology

The scraping process followed these steps:



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* Technologies & Libraries Used

Programming Language: Python Libraries

Requests: Used to send HTTP requests and retrieve web pages.

BeautifulSoup: Used to parse HTML and extract information.

Regular Expressions (re): Used to locate and extract values such as price, engine, mileage, and rating.

Pandas: Used for data collection, cleaning, manipulation, and analysis.

NumPy: Used for numerical operations and handling missing values.

Matplotlib: Used to create graphs and visualizations.

Jupyter Notebook / Anaconda: Used as the development environment.

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* Data Collection Process

Field	Description
Car Name	Name of Hyundai car
Brand	Hyundai
Price	Starting price in Lakh
Fuel Type	Petrol / Diesel / Electric
Transmission	Manual / Automatic
Engine	Engine capacity
Mileage	Reported mileage
Seating Capacity	Number of seats
User Rating	CarWale user rating

The collected information was first stored in:
hyundai_raw.csv

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* Raw Dataset Overview

Dataset Size
Rows: 56
Columns: 9

Raw Dataset Columns:

Car Name
Brand
Price
Fuel Type
Transmission
Engine
Mileage
Seating Capacity
User Rating

Raw Data File: hyundai_raw.csv
The raw dataset was preserved before cleaning so that the original scraped data remained available

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* Data Cleaning & Preprocessing

The raw dataset required several preprocessing steps:

Cleaning performed:

Removed "Lakh" from price values.
Converted Price into numeric format.
Extracted engine capacity into Engine_cc.
Extracted mileage values.
For mileage ranges, calculated an average mileage value.
Converted seating capacity into numeric values.
Converted user ratings into numeric values.
Changed "Not Available" into NaN.
Removed duplicate records.
Corrected invalid transmission values for electric cars.

Final cleaned file: hyundai_cleaned.csv

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* Exploratory Data Analysis

The main numerical results from the available data were:
Average Price: 13.89 Lakh
Average Mileage: 17.66 kmpl
Average Engine Capacity: 1247.36 cc
Average User Rating: 4.41

Fuel Type Distribution
Petrol — 10
Electric — 2
Diesel — 1

Price Category Distribution
Budget — 6
Mid-range — 6
Premium — 1

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* Key Analysis Questions / Use Cases

The project answers the following questions:

1. **What is the average Hyundai car price?**
2. **Which fuel type has the highest number of available cars?**
3. **How are Manual and Automatic transmission types distributed?**
4. **Which Hyundai cars have higher available mileage?**
5. **How does engine capacity vary between models?**
6. **Which cars have the highest available prices?**
7. **Which cars have the lowest available prices?**
8. **How do user ratings vary across models?**
9. **How many cars fall into Budget, Mid-range, and Premium categories?**
10. **What patterns can be observed between price, mileage, and rating?**

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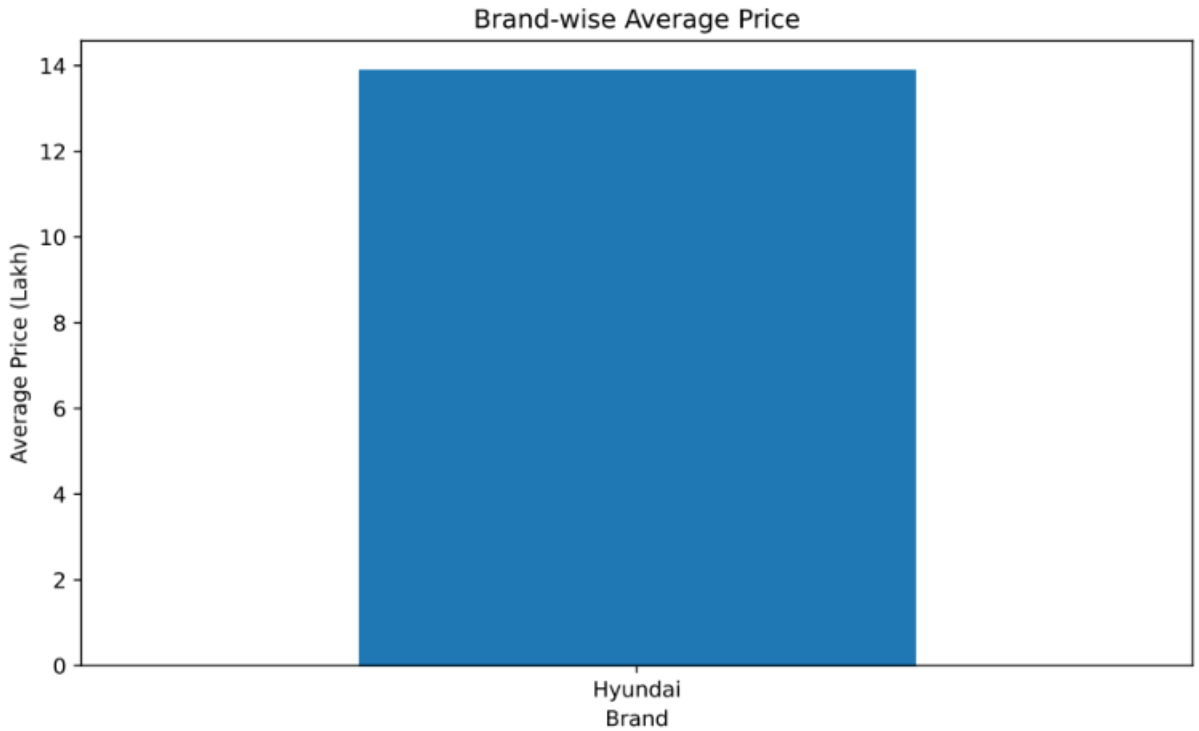
* Visualizations & Insights

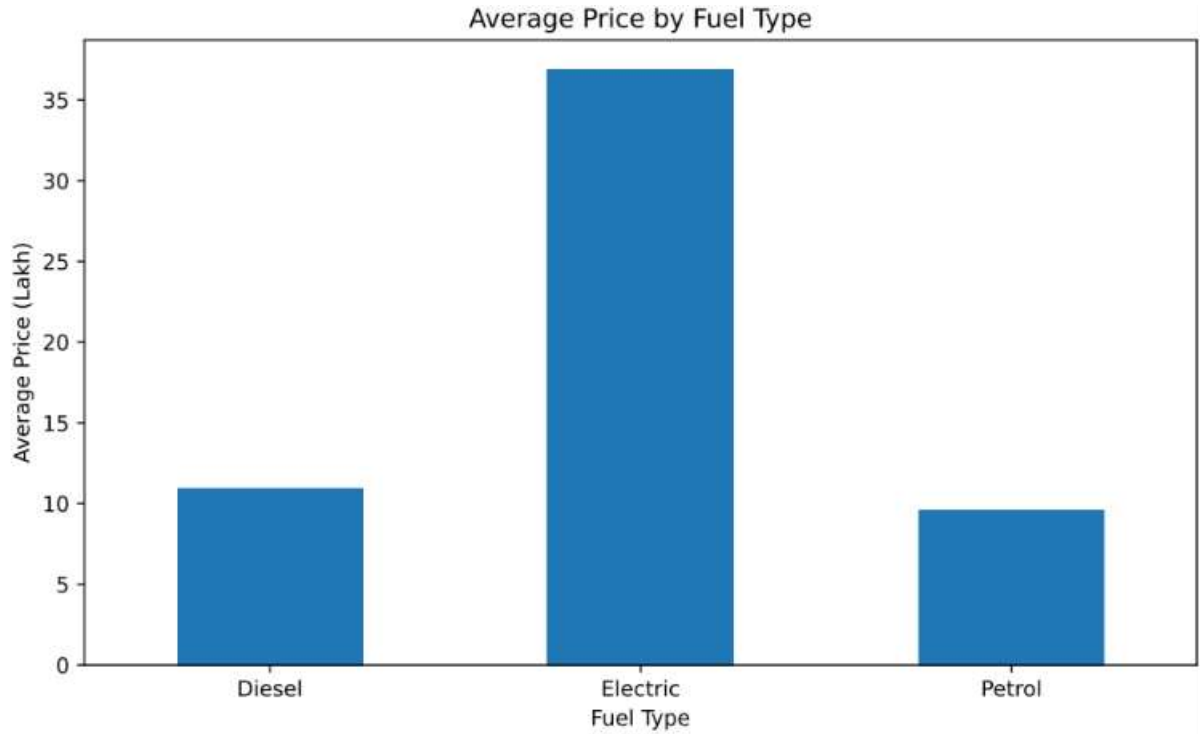
The project created the following visualizations:

Brand-wise Average Price
Average Price by Fuel Type
Transmission Comparison
Mileage Comparison
Engine Capacity Comparison
Most Expensive Hyundai Cars
Cheapest Hyundai Cars
User Rating Analysis
Budget / Mid-range / Premium
Fuel Type Pie Chart
Transmission Pie Chart
Price vs Mileage Scatter Plot
Price vs User Rating Scatter Plot

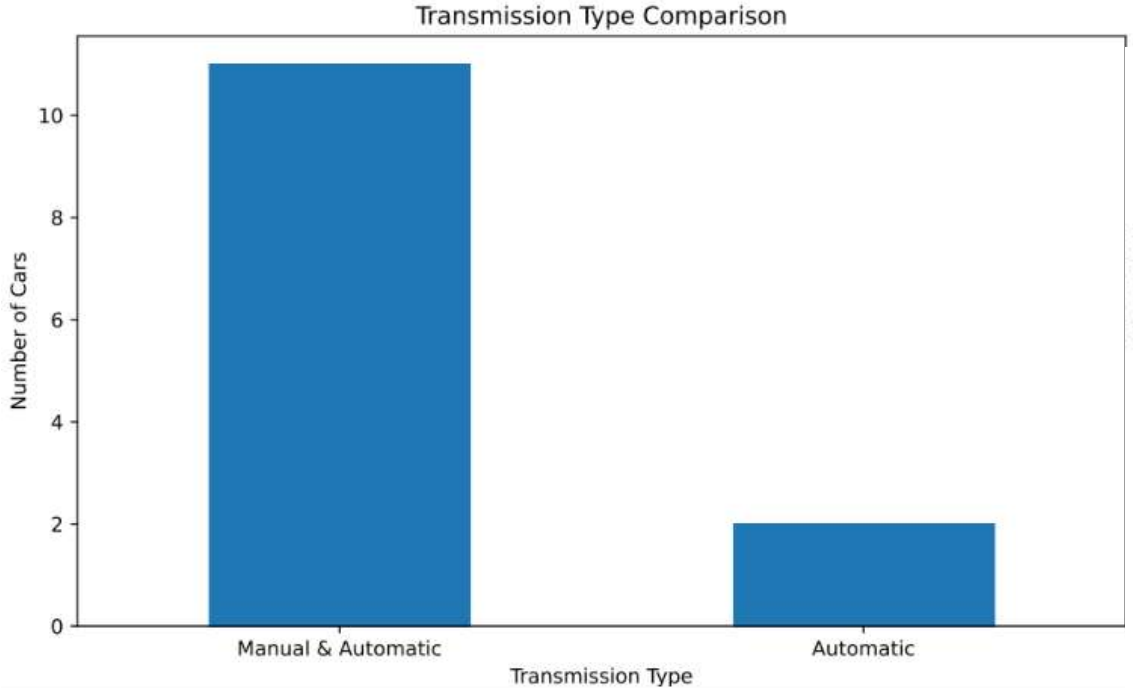
***Important observation**
Because this project uses **Hyundai-only data**, the brand-wise analysis represents Hyundai rather than comparing multiple automobile brands.

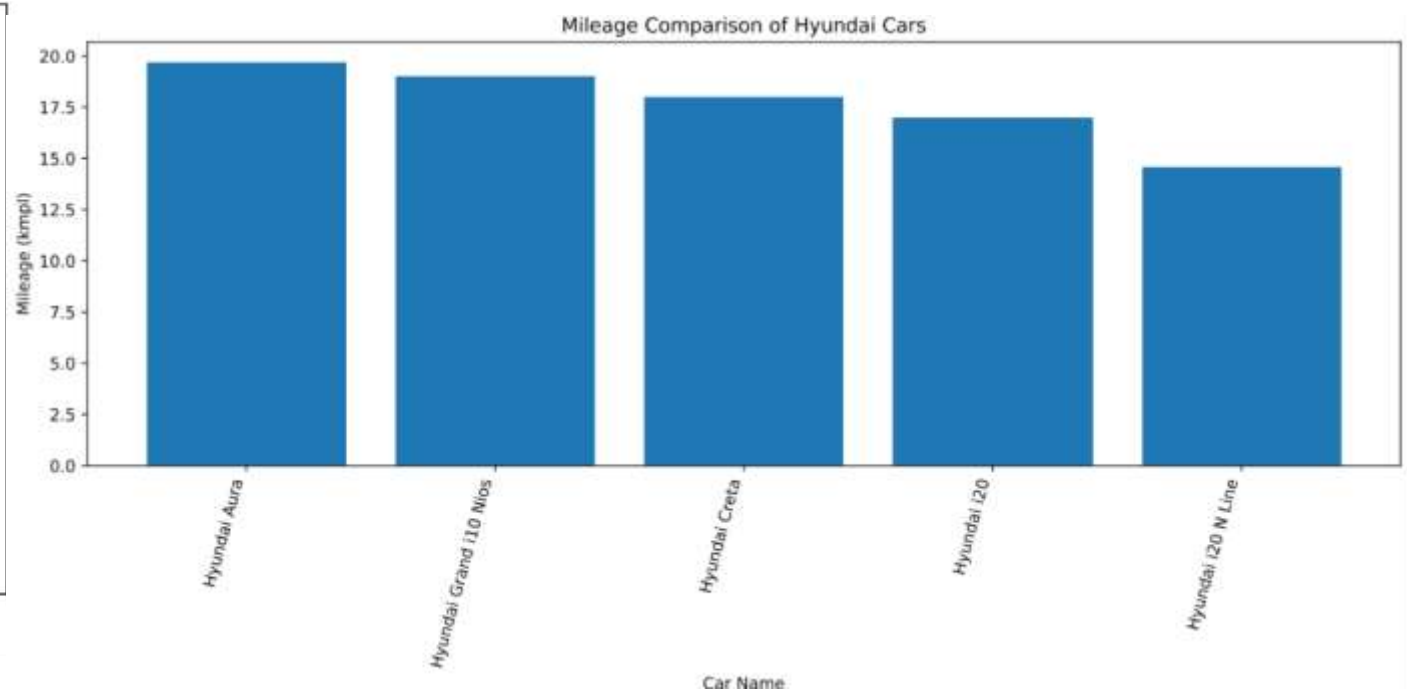
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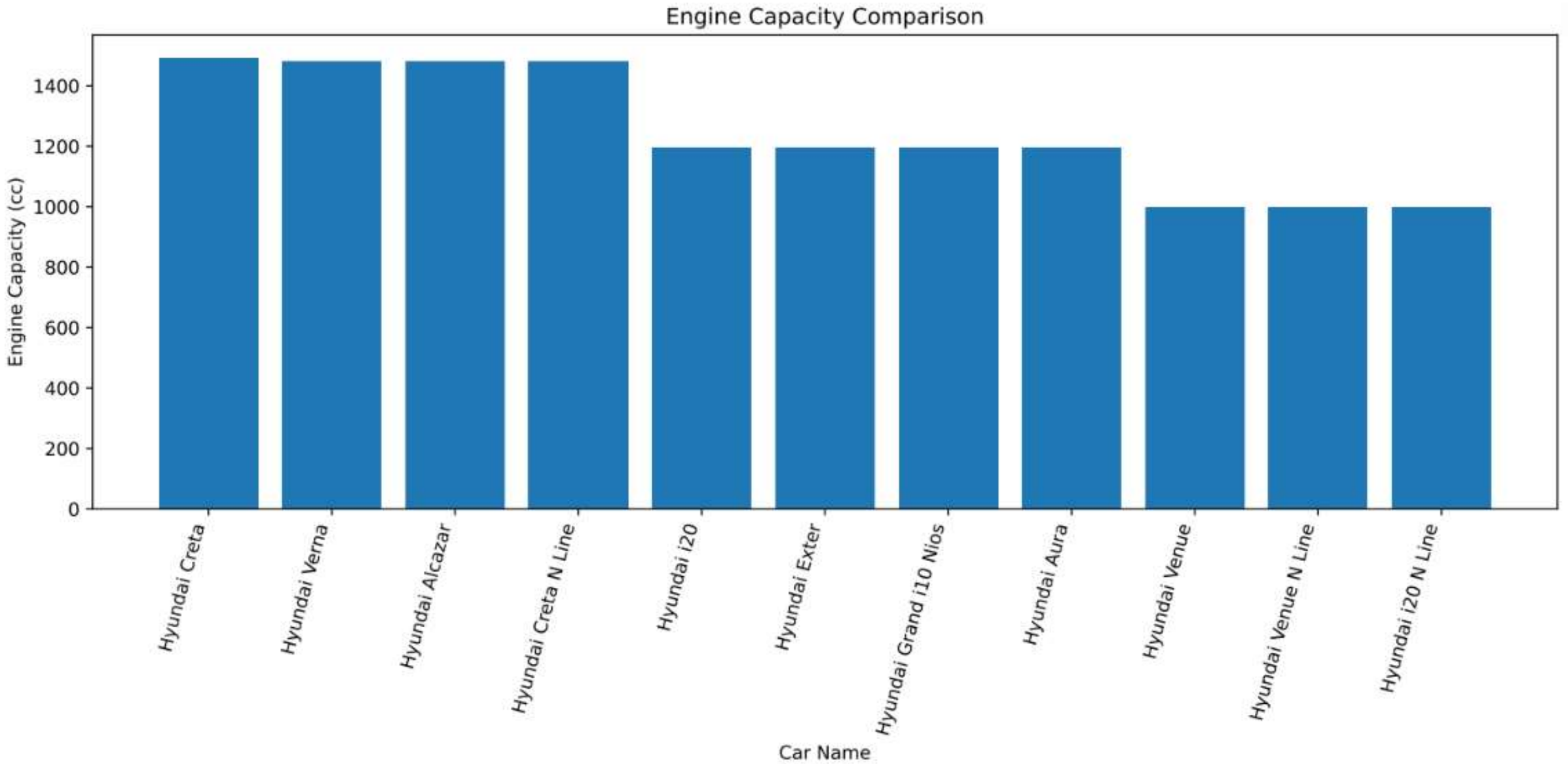


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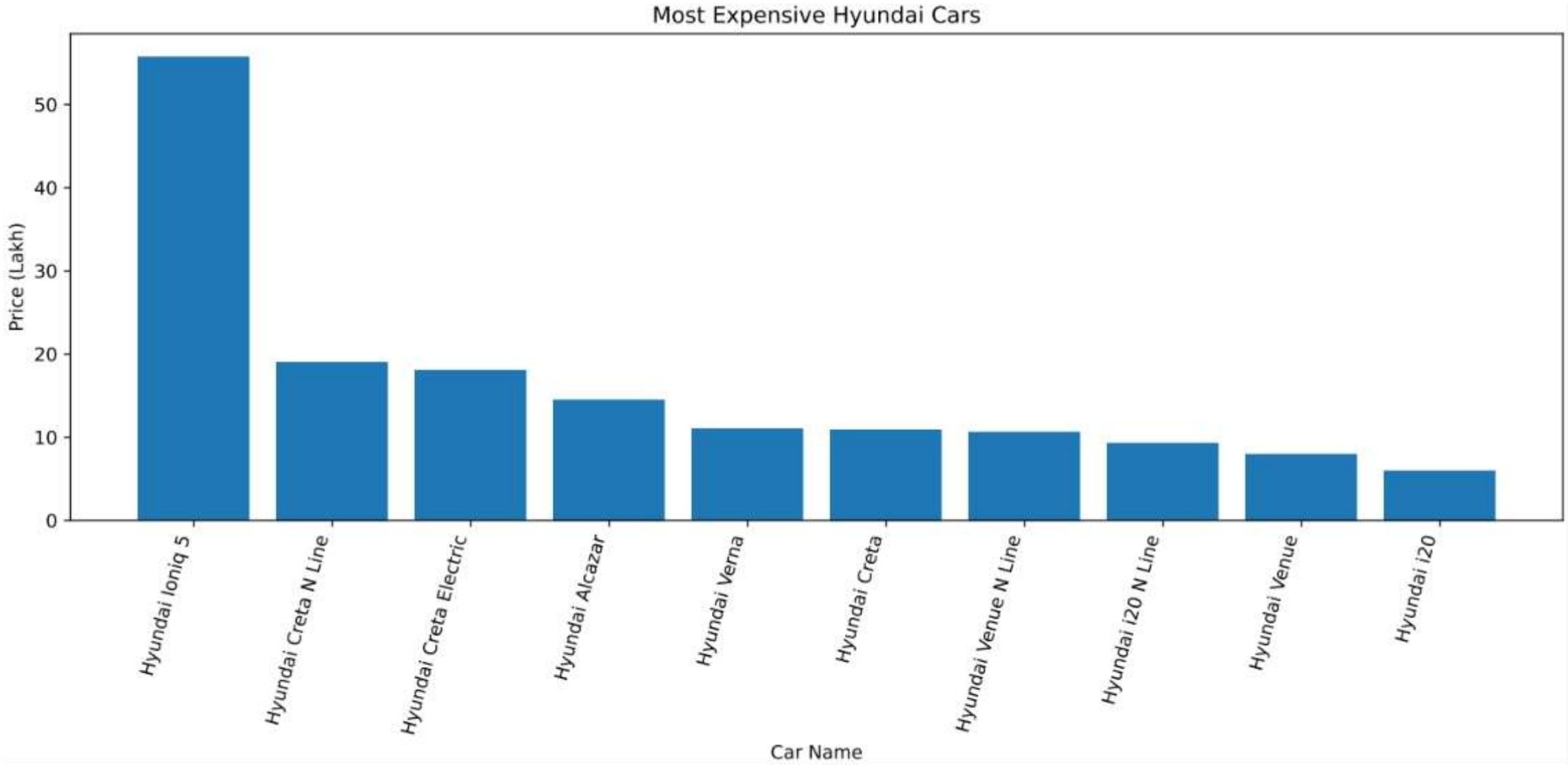




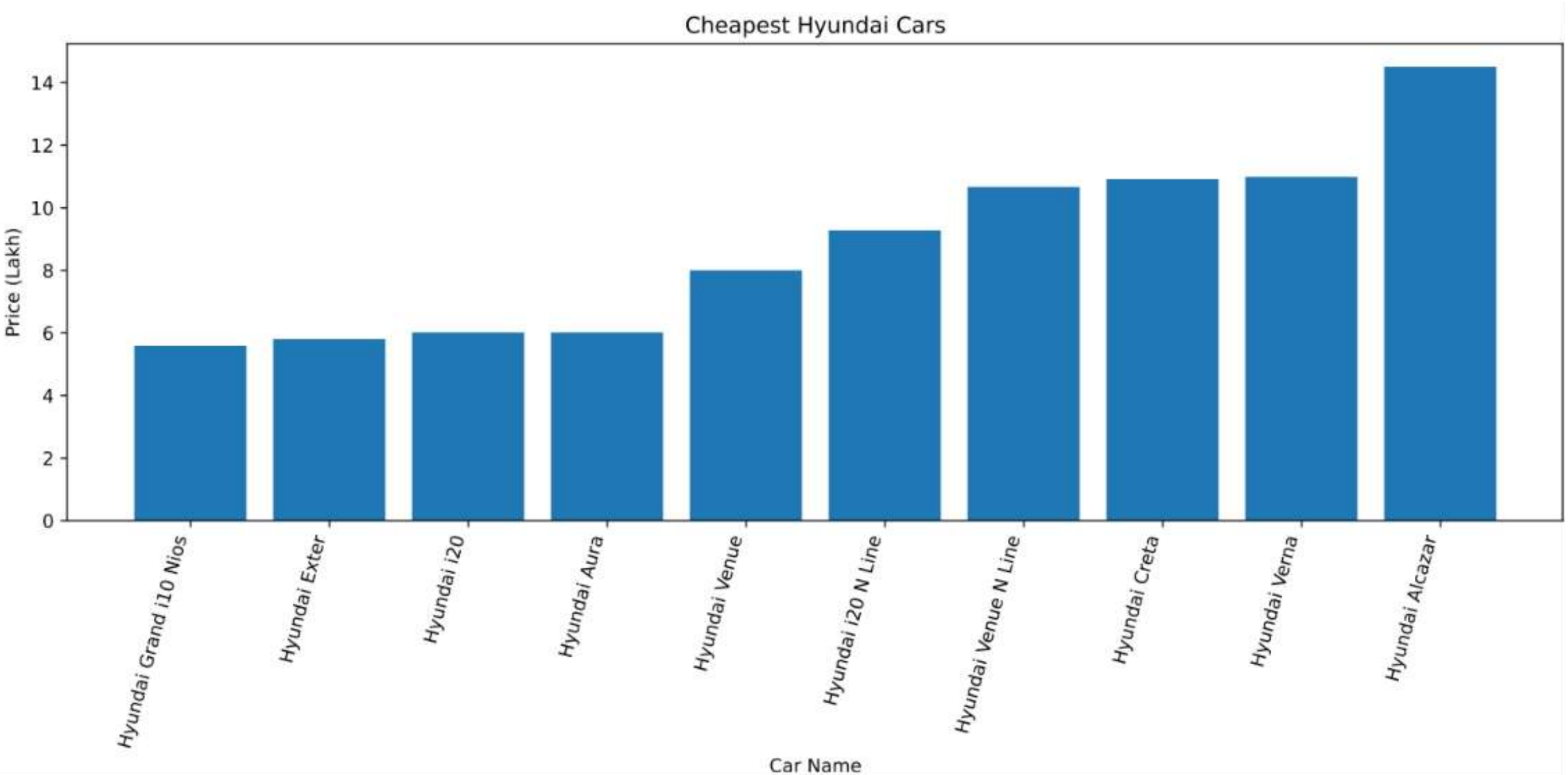
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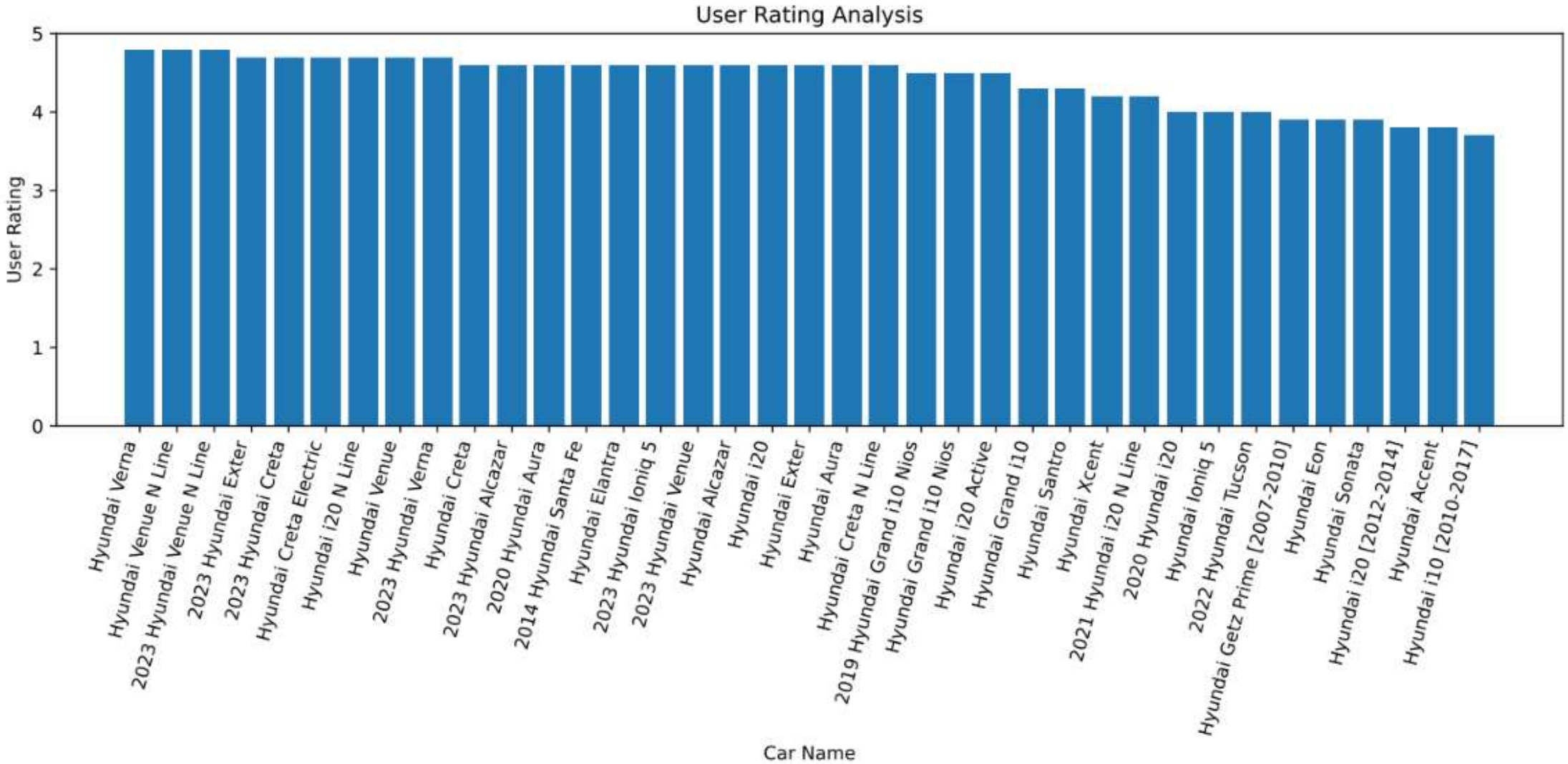
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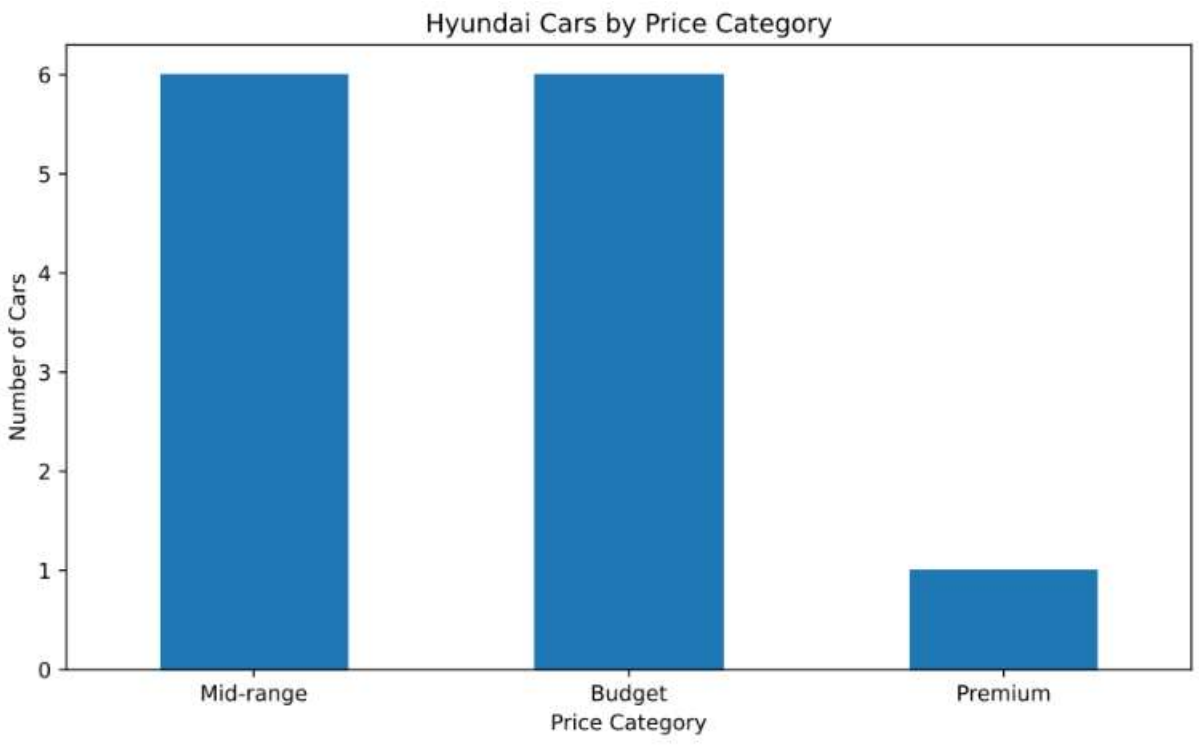
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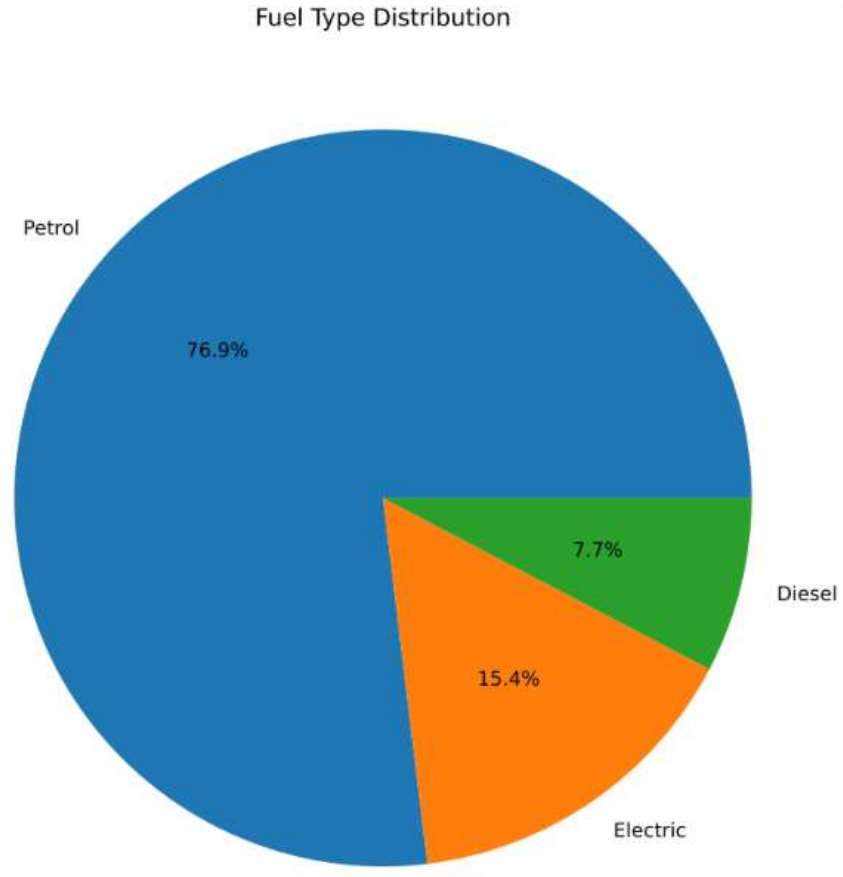


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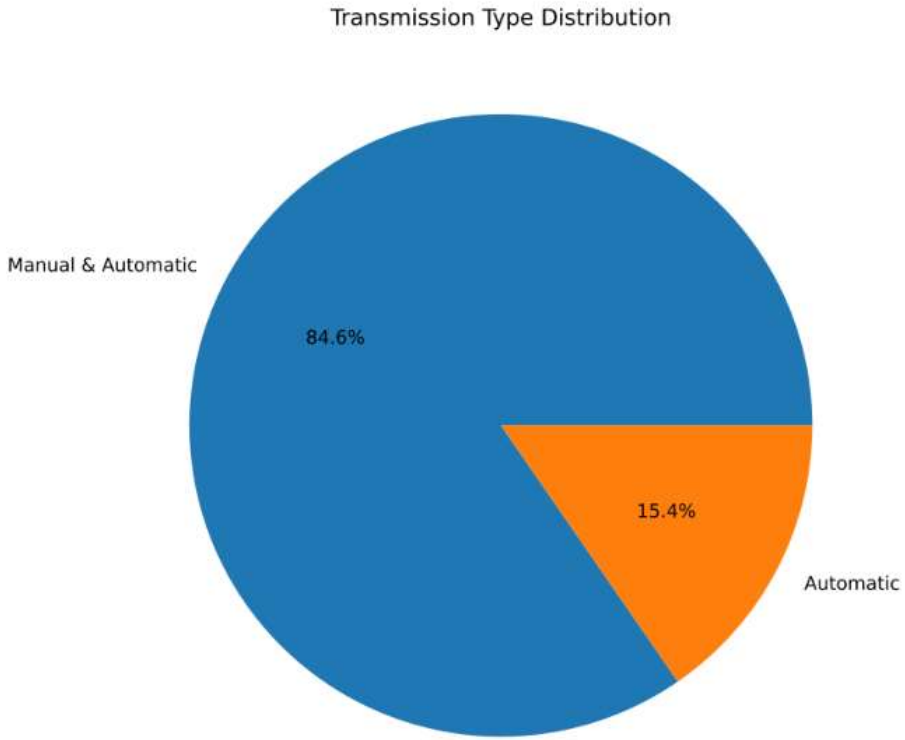


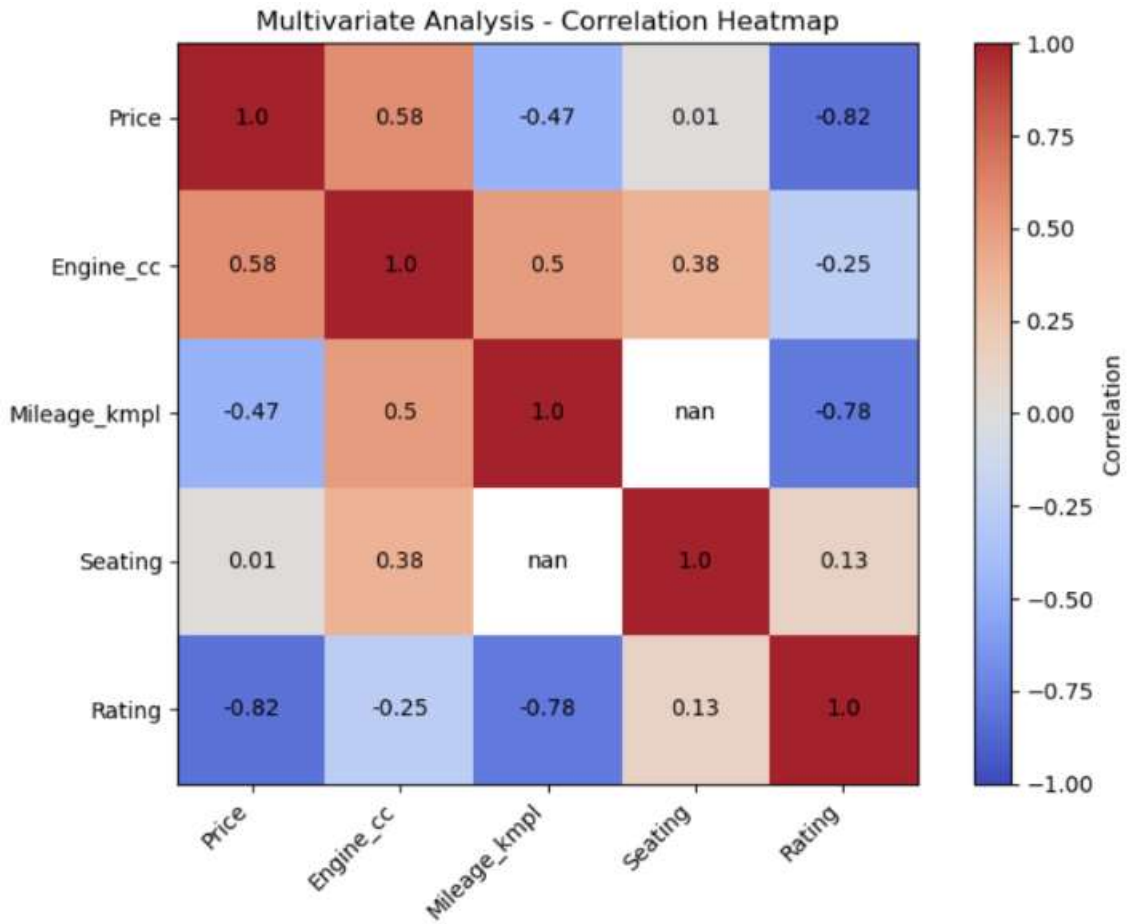
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* Conclusion & Key Findings

The project successfully demonstrated an end-to-end data analytics process using web scraping.

Key findings:
The average available Hyundai car price was **13.89 Lakh**.
The average available mileage was **17.66 kmpl**.
The average available engine capacity was **1247.36 cc**.
The average available user rating was **4.41**.

Petrol was the most common fuel type among the records with available fuel information.
The available priced cars were distributed into Budget, Mid-range, and Premium categories.
Visualizations made it easier to compare Hyundai models across different attributes.

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* Challenges Faced During the Project

Challenge 1 — Finding valid car URLs
The Hyundai page contained current, upcoming, and older model pages.

Challenge 2 — Extracting specifications
Different car pages presented information in slightly different formats.

Challenge 3 — Missing values
Some pages did not provide all required information in the same format.

Challenge 4 — Incorrect extraction
Some text, such as "Safety", was initially captured as transmission for electric cars.

Challenge 5 — Data cleaning
Price, mileage, engine capacity, seating capacity, and ratings needed to be converted into analysis-ready formats.

How I handled them
I tested individual pages, inspected the HTML/text structure, modified the extraction logic, validated the results, and then performed cleaning and analysis.

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* Future Improvements

Possible improvements for a larger version of the project:

1. Scrape multiple car brands instead of only Hyundai.
2. Collect variant-level information instead of one specification per model.
3. Add additional features such as horsepower, torque, battery capacity.
4. Automate the scraping process for multiple brands.
5. Build an interactive dashboard using Power BI or Tableau.
6. Add more advanced statistical analysis.
7. Schedule periodic scraping to monitor changes in the car market.
8. Deploy the analysis as a web application.

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THANK YOU



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